Frontend Development

HTML,CSS,JAVASCRIPT, REACT

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Types of Language

programming language

- programming languages are set of instructions or code which tells a computer what it needs to do
- Examples are Java, C, C++, C#
- Programming languages are most widely used to make software or driver

Scripting Language

- interpreted language , It runs or executes inside another program.
- giving the script to perform some certain task
- ***no scope of compiler*** in scripting languages. Scripting languages are most widely used to create a website.
- Examples are Javascript, PHP, Perl, Python

Markup Language

- prepare a structure for the data or prepare the look or design of a page
- doesn’t include any kind of logic or algorithm
- It tells the browser how to structure data for a specific page, layout, headings, title, table and all or styling a page in a particular way. So basically it involves formatting data or it controls the presentation of data
- Examples are HTML or XML

HTML, CSS, JAVASCRIPT

HTML

- Hyper Text Markup Language
- describes the structure of a Web page
- consists of a series of elements & tell the browser how to display the content

```html
<!DOCTYPE html>
<html>
<head>
<title>Webpage</title>
</head>
<body>

<h1>Heading</h1>
<p>Paragraph</p>

</body>
</html>
```

Tags

- <h1>
- <p style="color:red;" > red paragraph.</p>
-
- Link
- <img width="50px" height="100px" src="https://t4.ftcdn.net/jpg/01/43/42/83/360_F_143428338_gcxw3Jcd0tJpkvvb53pfEztwtU9sxsgT.jpg" alt="Person image" >

- **HTML Formatting Elements**
  - `<b>` - Bold text
  - `<strong>` - Important text
  - `<i>` - Italic text
  - `<em>` - Emphasized text
  - `<mark>` - Marked text

Table

- arrange data into rows and columns.
- describes the structure of a Web page
- consists of a series of elements & tell the browser how to display the content
- Example below with Style, colspan, rowspan

```html
<table style="width:100%" >
  <tr>
    <th style="width:70%" >Company</th>
    <th colspan="2"> Contact</th>
    <th>Country</th>
  </tr>
  <tr style="height:200px" >
    <td >Alfreds Futterkiste</td>
    <td>Maria </td>
    <td>Anders</td>
    <td>Germany</td>
  </tr>
  <tr>
    <td>Centro comercial Moctezuma</td>
    <td>Francisco </td>
    <td>Chang</td>
    <td>Mexico</td>
  </tr>
  <tr >
   <td rowspan="2">Francisco </td>
    <td>Chang</td>
    <td>Mexico</td>
    <td>spanned row</td>

  </tr>

  <tr>

   <td>Francisco </td>
    <td>Chang</td>
    <td>Mexico</td>
  </tr>
</table>
```

List

- group a set of related items in lists.
- unordered HTML list , ordered HTML list
- description list is a list of terms, with a description of each term <dl></dl>

```
<ol type="1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

<ul style="list-style-type:circle;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

Type of Element

- Block elements - A block-level element always starts on a new line, and the browsers automatically add some space (a margin) before and after the element

- commonly used block elements are <p> and <div>

- Inline element - does not start on a new line , only takes up as much width as necessary. Example <span>

```
<div>
  <p>This is an inline span <span style="border: 1px solid black">Hello World</span> element inside a paragraph.</p>
</div>
```

CSS

- Cascading Style Sheets
- Describes how HTML elements are to be displayed on screen, use to style a Web page
- `selector` - points to the HTML element , Each declaration includes a CSS property name and a value, separated by a colon.

```
p {
  color: red;
  text-align: center;
}

.para {
 color: blue;
}

#para {
    color: white;
}

h1, h2, p {
  text-align: center;
  color: red;
}
```

Three Ways to Insert CSS

- External CSS
- Internal CSS
- Inline CSS

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="mystyle.css">
</head>
<style>
.customHeading {
      margin-left: 40px;
}

</style>

<body>

<h1 class="customHeading">My Heading</h1>
<p >New paragraph </p>

<div style="color:red;"> This is a paragraph </p>

</body>
</html>

mystyle.css –>

h1{
      color : red
}
```

Some Properties & Values

- `background-color` property specifies the background color of an element
- `border-style` properties allow you to specify the style, width, and color of an element's border
- `margin` - CSS margin properties are used to create space around elements, outside of any defined borders
- `padding` - Padding is used to create space around an element's content, inside of any defined borders

Javascript

- program the behavior of web pages
- getElementById() - Document Object Model
- document , window, console

```
window.location.pathname
window.location.assign(“https://www.google.com/")
```

- <script> Tag used to write the Javascript code in Html , If External .js file used

```
<script src="myScript.js"></script>
```

```
document.getElementById('myElement').innerHTML = 'Hello JavaScript';
document.getElementById("myElement").style.fontSize = “35px";
document.getElementById("myElement").style.display = “none”;
```

JavaScript Variables

- var - Always declare variables , support old browsers , can be redeclared
- let - block scope , must be **Declared** before use, cannot be **Redeclared** in the same scope
- const - if the value should not be changed , if the type should not be changed (Arrays and Objects) , then use this

```
const PI = 3.141592653589793;

var x = 5;
let y = 6;
let z = x + y;

let person = "John Doe", carName = "Volvo", price = 200;
```

JavaScript Concepts

- Operators (Arithmetic Operations , Assignment Operators)
- Datatypes (String, Number , Bigint ,Boolean , Undefined , Null
  - Object (An object, An array, A date)

Operators :
x = x + y
Assignment :
x += y

Datatypes :

```
let length = 16;
const person = {name:"Raja", location:"Chennai"};
const marks = [90, 56,88];
const date = new Date("2022-03-25");
```

JavaScript Concepts

- Functions - block of code designed to perform a particular task, reuse code
- JavaScript Objects (**name:values** pairs in JavaScript objects are called **properties)**
- Array of objects (Consist set of objects)

```
function name(parameter1, parameter2, parameter3) {
  // code to be executed
}


const person = {
  firstName: "John",
  lastName : "Doe",
  id       : 5566,
  fullName : function() {
    return this.firstName + " " + this.lastName;
  }
};

Let arrayObj = [
{
     Name : “Raja”,
     Subject : “English”
},

{
     Name : “Mani”,
     Subject : “Maths”

}
]
```

String Functions

- Extracting String Characters
- String Search

```
let text = "mystring";
let char = text.charCodeAt(0);

let text = "Apple, Banana, Kiwi";
let part = text.slice(7,13);
let part = text(7, 13);

text.toUpperCase();
text.concat(" “, “ tostring”);
```

Thank you